Capstone Project

AMAes Research Manual

I. Introduction

- In recognition of research's integral role in an educational institution, AMAES consistently gears toward the improvement of the institution's research productivity, evidenced by its strong support to encourage and allow the members of the academe – faculty members and the students – to produce scholarly output in and out of AMAES.

Intro…

- At AMAES, the contribution of students to the research-effort is considered essential for the development of researchoriented and research-centered institution. Original and innovative research study that makes significant contribution in knowledge building is continuously recognized.

Intro…

- Thus, to continue its principle, this Policy Manual on Research Writing is developed to assist the students in writing research project – thesis, design project, capstone, dissertation, business plan, feasibility and case study – for the fulfillment of their academic requirements. It provides student with detailed research guidelines and procedures for ready reference.
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II. Engaging Students to Conduct Research

- In a learning-centered environment, research and inquiry is inextricably linked together. This is where a discovery-driven attitude of the students is built away from just a passive repository of transmitted information from the professors. This is where research becomes an integral part of every student’s academic life.

Engaging Students to Conduct Research

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- By engaging in research, students could develop analytical and critical thinking and their writing skills could be trained. These are the skills that are essential to every professional individual in and out of the academe, and more in the industry.

III. Vision

- AMA Education System envisions being a global leading center for research on Computer Science and Information Technology, while being among the leading research centers on engineering, business administration, and accountancy. The institution will help translate academic knowledge into workable tools for success and development in the private and public sector.

IV. Mission

- It is the mission of AMA Education System to generate, synthesize, and transmit knowledge for and about engineering and information technology resulting in quality learning. It serves as a catalyst in assisting scholars, educators, and practitioners to develop research, teaching, and leadership capabilities in school and non-school settings.
- The different brands and colleges under AMA Education System umbrella are charged with the development of highend, research-oriented faculty and students, encouraging creative endeavors and intellectualism, recognizing research talents of the highest distinction by providing incentives for scholarly achievements.

V. Program Goals

- In line with the AMA Education System's vision and mission, the following objectives will be pursued:

1. To promote and support new and innovative applied and developmental researches in the fields of General Education, Humanities and Social Sciences, Information Technology, Engineering, Business Administration and Accountancy.
2. To disseminate research outputs through publications and presentation in national and international conferences, seminars, symposia or fora.

Program Goals

3. To establish and maintain linkages – partnership and collaborations – with leading educational institutions, government, industry and other related organization in strengthening its research efforts.

4. To continuously nurture and enhance the growth and development of research across AMAES.

VI. Research Agenda

- AMA Education System's research agenda is aligned with the research policies, directions, priorities, and thrusts set by the National Higher Education Research Agenda (NHERA):

1. Research is the ultimate expression of an individual’s innovative and creative powers. The higher education sector shall ensure that the academic environment nurtures and supports Filipino research talents.

Research Agenda

2. Research thrives in an environment characterized by free flow of information, honest, and analytical exchange of ideas, and supportive policy and administrative structures. Higher education policies

shall enhance Institution’s and the individual’s capacity to conduct independent, collaborative and productive research.

Research Agenda

3. Research is one of the main functions of the higher education sector. Universities in particular, are expected to lead in the conduct of discipline-based, policy-oriented, technology-directed and innovative/ creative researches that are locally responsive and globally competitive.

Research Agenda

In line with these research principles, CHED identified priority research areas such as family and community transitions, education, disaster, ICT and social change, energy, health and food security. AMAES, on the other hand, align its research priorities in the following areas:

- Enabling Research Areas
- Applied and Multi-disciplinary
- Technology or Theory Discovered and Developed
- Economic and Social Benefits
- Environmental Engineering
- Food and Security

Research Agenda

Priority research areas include, but are not limited, to the following:

Computer Studies

1. Cloud Computing
2. Cybersecurity
3. Database and Data Management
4. Data Mining and Analytics
5. Computer Graphics
6. Computer Architecture
7. Artificial Intelligence Technology
8. System and Software Development
9. Networking
10. Mobile Computing

VII. Expected Research Output Per Program

Research is part of the Bachelor, Master, and Doctor's Curricula. This is linked together in response to an academic assignment and in conducting major research project -thesis, design project, capstone, dissertation, business plan, feasibility and case study. This is a requirement for the completion of an undergraduate and graduate degree program:

Program	Type of Research Output
Bachelor of Science in Computer Science	CS Thesis Writing 1 & 2
Bachelor of Science in Information Systems	IS Capstone Project 1 & 2
Bachelor of Science in Information Technology	IT Capstone Project 1 & 2

VIII. Guidelines for Student Researches

RULES FOR UNDERGRADUATE RESEARCH WRITING

- When a student reaches senior standing, he/she becomes eligible to enroll in Thesis A.
- Only upon successful completion of Thesis A (or NFE mark has been removed) will the student be allowed to enroll in Thesis B.
- Student(s) work hand-in-hand with the thesis adviser for the completion and finalization of his approved capstone/thesis/design project.
- Students from the same college may form a group of at most three (3) members to conduct one specific research.

Guidelines for Student Researches

- Change of topic and group members will not be allowed once the research proposal was submitted, defended and approved.
- Multi-disciplinary research or design projects will be subjected for approval of the Academic Affairs Department.
- For meritorious cases such as size and coverage of the project, the number of group members maybe increased to at most 5 students. A letter of appeal on the number of group members, stating reasons for said appeal must be prepared. Head of Academic Affairs and Operations Department must approve the letter of appeal.

IX. Defense Process

Undergraduate Programs:

Defense Process	Course Offering
I. Proposal Presentation	Methods of Research
II. Pre-Oral Defense	Thesis A / Design Project 1 / Feasibility Study 1
III. Final Oral Defense	Thesis B / Design Project 2 / Feasibility Study 2

X. Parts of a Research Paper Per Program Output

2. Outline for BS Information Technology, and BS Information Systems Capstone Projects (CMO No. 25 Series of 2015)

Preliminaries	Title Page Approval Sheet Acknowledgements Abstract Table of Contents List of Tables List of Figures List of Appendices List of Abbreviations and Symbols (Optional) Definition of Terms
Chapter 1	**PROJECT AND ITS BACKGROUND**
	Project Context Purpose and Description Objective of the Study Significance of the Study Scope and Limitation
Chapter 2	**RELATED LITERATURE**
	Foreign Literature Local Literature Foreign Studies Local Studies Synthesis Technical Background
Chapter 3	**METHODOLOGY RESULTS AND DISCUSSION**
	Software Design, Products and/or Processes System Architecture Conceptual Design Cost Benefit Analysis Requirement Analysis System Architecture/System Flow Block Diagrams Development and Testing Input and Output Reports and Analysis Description of the Prototype Implementation Plan Implementation Results (optional)

Chapter 4	**CONCLUSIONS AND RECOMMENDATIONS**
	This chapter includes a summary of the main findings of the study. It also presents the significance of the study, and relates findings to the objectives and problems written in the introduction part of the study. Recommendation/s must be stated in this chapter. This part usually directs the reader to conduct further research on some specific areas related to the study.
References	This is a list of works cited, as well as works consulted but not cited (example, background reading not necessarily cited) in the construction of the research paper. The list of references is numbered and arranged alphabetically. For format on the writing of references, the IEEE (Institute of Electrical and Electronics Engineers) style of documentation shall be followed.
Appendices	An appendix or appendices, if any, should be after the bibliography. Appendices include original data, preliminary tests, tabulations, tables that contain data of lesser importance, very lengthy quotations, forms and documents, computer printouts and other pertinent documents. Appendices should be arranged chronologically as they are cited in the main text. Use capital letters of the English alphabet to track appendices, and always begin with the letter A.

XI. Manuscript Preparation

1. **Paper Specification**

- Use white letter-size paper (8 ½ x 11 in).
- Orientation should be portrait style.
- Text and figures are placed on only one side of the paper.
- The other side should be left blank.

Manuscript Preparation

2. **Font Type and Size**

   - AMAES uses American Psychological Association (APA) 6th edition (for non-technological programs like Business, Arts, Humanities and Social Sciences and Education, and Institute of Electrical and Electronics Engineers (IEEE, for technological programs like the DIT, MIT, MSCS, BSIT, BSCS, BSECE, BSCpE, BSIE) style of writing and its official font type and size varies from what style is using. IEEE uses Times New Roman with a size of 10pts and APA 6th Edition uses Arial with a size of 12 pts.

Manuscript Preparation

3. Pagination

- Pages should be numbered consecutively throughout the research paper, including pages of figures, tables, appendices and bibliography.
- Page 1 begins with the first page of the first chapter.
- Preliminary pages preceding the first chapter are numbered using small Roman Numerals (i, ii, iii, etc). The title page should not be numbered although it is counted as page i.
- Page numbers are placed at the center of the bottom of each page.

Manuscript Preparation

4. **Spacing of text**

- The body of the research paper should be typed double-spaced. The following, however, should be single-spaced:
- abstract
- footnotes
- quotations longer than three lines
- references and bibliography (except between entries)
- headings or subheadings
- multi-line captions of tables, figures, illustrations, etc.

5. **Text alignment and Paragraph Indentation**

- The text should be justified while keeping the first line of every paragraph indented.

Manuscript Preparation

6. **Margin**

- For the first page of any chapter, the left, top and bottom margins should be 40mm wide.
- For the rest of the pages, the left margin should be 40mm and the top and bottom margins should be 25mm.
- The right margin is 25mm for all pages. Make sure text is always justified on both sides. Use a five-space indention for every paragraph.
- Margin specifications are meant to facilitate binding and trimming. All information (text headings, footnotes, illustrations, etc), including page numbers must be within the text area.

Manuscript Preparation

7. Tables and Figures labeling

- Type above every table its number and title.
- Type the table number at the center and triple-spaced from the last line of the immediately preceding paragraph.
- Type the title of the table at the center, title case, and double–space from the table number, and triple-spaced to the table.
- Type triple–spaced from the last line of the immediately preceding paragraph its figure number and title.
- Type the title of the figure flush left, title case, and single–space immediately after the figure number.
- The numbering of tables and figures shall be continuous.
- If the title is long, align to the first word of the title the rest of the title.

Manuscript Preparation

the Arial font to a minimum of 8 points.

- If the table does not fit the page even after the font-size reduction, continue the rest of its part to the next page, but the cut part shall bear the subtitle, ‘Continuation’flush with the left margin (e.g., Continuation of Table 14). The cut part/s shall also show the column headings and the legend (if any) like those in the first part of the table.
- Put two or more small tables or figures on a single page if the page is big enough to accommodate them all.
- Have small tables and figures appear on the same page along with the text; however, leave three spaces between every table/figure and the texts above and below it.
- The width of the table shall be dependent on the length of its title.

XII. Binding Specifications and Submission of Manuscripts

After the student/s successfully defended the thesis, integrated all recommendations and revisions, obtained all necessary signatures the manuscript must be photocopied and hard bounded.

Students must obtain at least five (5) hardbound copies of their approved thesis: Copy for library, respective college, research office, adviser's copy, personal copy.

The following should be lettered in gold from the head to the foot of the thesis spine, using a 24-point font: (1) proponent/s; (2) degree for which the work is submitted; (3) research title; and (4) year of submission.

Manuscript Preparation

Table 1
Competitive Advantage of Electronic Whiteboard Eraser

Feature	Traditional Whiteboard	Touchscreen LED	Electronic Whiteboard Eraser
Price	cheapest	most expensive	relatively expensive
Writing tool	marker	electronic software	marker
Erasing tool	manual eraser	electronically	motorized eraser
Automated?	No	Yes	Yes
Needs power?	No	Yes	Uses power supply
Saves time?	No	Somewhat	Yes
With timer?	No	Depends on model	Yes

- Use the landscape page layout for large tables.
- If the table is still too large to fit a single page, reduce

Binding Specifications and Submission of Manuscripts

The following particulars should be provided on the thesis cover, using 18-point gold block font: title of thesis; proponent/s; degree; name of school; and year of submission. The layout of the cover is very similar to that of the title page.

Following should be the cover color of the bound thesis:

College of Arts and Sciences	-	Maroon
College of Business Studies	-	Midnight Blue
College of Computer Studies	-	Royal Blue
College of Engineering	-	Dark Green
School of Graduate Studies	-	Black

CD Label. For master thesis, dissertation, capstone project and design project, the CD case cover follows the format of the Title Page on the hardbound copy.

XIII. Research Writing Convention

1. **Language**

   - The official language to be used in writing a research paper is American English.
   - Use only objective, expository English. Idioms, slang, subjective words and phrases, metaphors and undefined acronyms are not acceptable.
   - All sentences must be complete with acceptable grammar and correct spelling.

Research Writing Convention

3. Units of Measure

- Whenever applicable, use SI units of measurement.
- Always use internationally recognized abbreviations for units of measure, and do not put a period after them. For example: 20 milliliters (20ml); 2.5 hectares (2.5ha); 3.7 metric ton (3.7mt) 80 kilometers per hour (80kph).

Research Writing Convention

2. Numbers

- Spell out all numbers less than ten unless they are attached to units of measurement (e.g. 5kg, 10ml).
- Use figures for 10 or numbers more than 10.
- If a sentence begins with a number, write the number in words even if it is more than 10.
- If a series of figures is to be used, use numerals regardless of the value. Example:
- In the room were 4 chairs, 12 boxes, 13 books, 10 files, 9 umbrellas and 8 pairs of shoes.
- The number of computer units bought during the last five years was 8, 53, 27, 38, 52 and 9.

Research Writing Convention

4. **Direct Quotations**

- Quotations must be copied accurately, word for word, and they must be placed in quotation marks unless they have been formally set off from the rest of the text.
- Parenthetical citation must follow each direct quote. The page number where the quote appears is included, in case the quote is lifted from a book. Otherwise, follow APA standards.
- Direct quotations must be minimized at all costs, except in some fields such as literature. Extensive use of direct quotations can be quite irritating for the reader.
- Students must learn to synthesize and paraphrase concepts in their own words and style.

Research Writing Convention

5. Footnotes

- Footnotes should be used sparingly regardless of the field of study. However, it is recommended that citations be used instead of footnotes. Extensive footnoting tends to distract the reader from the main argument of the text.
- In cases where footnotes are necessary, the indicators used are superscript. The numbering of footnotes should begin with 1 and must be continuous throughout the whole text.

Research Writing Convention

6. **Grammatical Tenses**

- As a rule of thumb, use the present tense when referring to previously published work and the past tense when referring to present results of the study.
- In a thesis, it is normal to go back and forth between present and past tenses. Most of the abstract should be in the past tense because present results are being described.
- On the other hand, the introduction, discussion and conclusion should be in the present tense because these chapters usually refer to previously published works.

XIV. Post-Final Defense Process

After a student successfully defended his/her final project, a certification of final version of capstone/thesis/dissertation shall be prepared along with the following requirements:

1. Hardbound copy of manuscript
2. Electronic submission saved in CD-ROM (includes the following: Journal article, code, exe files, user manual, CV and manuscript)
3. Journal Article (only for graduate students)
4. Anti-Plagiarism Test Result not more that 20% similarity index
5. Certificate of Proofread
6. Curriculum Vitae

XV. Defense Fees

- Panel – Php **1500 per student**– must be a full pledge Master
- Adviser – Php **1000 per group** – must be in the field BSIT/BSCS or related.
- Grammarian – Php **1000 per group** – can be an English major with experience in research writing/ can be in the field BSIT/CS, or related, with experience in proofreading.

XVI. Grading system

Grade Range	Grade Point	Grade Input	Descriptive Rating
96 – 100	1	A+	Excellent
91 – 95	1.25	A	Very Good
86 – 90	1.5	A-	Very Good
81 – 85	1.75	B+	Good
75 – 80	2	B	Good
69 – 74	2.25	B-	Good
63 – 68	2.5	C+	Fair
57 – 62	2.75	C	Fair
50 - 56	3	C-	Fair
Below 50	5	F	Failed
	IP	IP	In Progress

Grading system

For capstone project, panel shall evaluate according to the following criteria:

Rating Criteria	Weight
FINAL PAPER GRADE (30%)	
Originality/Inventiveness/Scientific Impact	15%
Quality of Manuscript	15%
PROTOTYPE/SOFTWARE GRADE (35%)	
Conceptual & Logical Design	20%
Workability & Application	15%
ORAL PRESENTATION (35%)	
Quality of Presentation	20%
Ability to Defend Thesis/Dissertation	15%
Total	**100%**

Capstone Project 1 Notable Schedules

- **Schedules for checking outputs**

September 19-20 Prelim

October 10-11 Midterm

October 24-25 Prefinal

November 21-22 Final

- **Oral Defense schedule:**

December 5, 2025 (WEEKDAYS) 8 AM-5 PM F2F VENUE ACLC Conference room

December 6, 2025 (WEEKENDS) 8 AM-5 PM F2F VENUE ACLC Conference room

- Requirements
  - Hard copy of the complete Chapters 1, 2, and 3 and List of System Features
    - It must be signed by the adviser
    - It must have already been proofread
  - Business attire.
  - Panel Fee, adviser fee, and grammarian fee acknowledgment Receipts from the accounting office.